if the picture were made much larger, you would see tertiary and higher-order images, closer and closer to the shadow.

Can you figure out why the lensed disk has the form you see? Why is the primary image wrapping under the shadow attached to the thin secondary image wrapping over it? Why are the paint swatches on the over-wrapping and under-wrapping images widened so greatly, and those on the sides squeezed? . . .

Gargantua's space whirl (space moving toward us on the left and away on the right) distorts the disk images. It pushes the disk away from the shadow on the left and toward the shadow on the right, so the disk looks a bit lopsided. (Can you explain why?)

To get further insight, Eugénie von Tunzelmann and her team replaced their variant of the color-swatch disk (Figure 9.7) with a more realistic thin accretion disk: Figure 9.9. This was much more beautiful, but it raised problems. Chris did not want his mass audience to be confused by the lopsidedness of the disk and black-hole shadow, and the shadow's flat left edge, and the complicated star-field patterns near that edge (discussed in Chapter 8). So he and Paul slowed Gargantua's spin to 0.6 of the maximum, making these weirdnesses more modest. (Eugénie had already omitted the Doppler shift caused by the disk's motion toward us on the left and away on the right. It would have made the disk far more lopsided: bright blue on the left and dim red on the right—totally confusing to a mass audience!)